EKF Mathematical Formulation

sensor_fusion documentation

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Contents

1 Scope

This document describes the runtime EKF implemented by `sensor_fusion/src/ekf/` and the included generated model files. The filter estimates the NED-frame attitude with respect to the vehicle frame, velocity, position, IMU biases, and the physical vehicle-to-body mount used during inertial propagation. It is the primary runtime inertial/GNSS filter behind `SensorFusion`.

2 Frames and Mount Sources

Symbol	**Meaning**
n	Local NED frame. EKF position and velocity states are in this frame.
b	Raw IMU body frame.
v	Physical vehicle frame, FRD: forward, right, down.

Rotations follow

$$x_a = C_{ab} x_b, \qquad R(q_{ab}) = C_{ab}, \qquad R(q_1 \otimes q_2) = R(q_1) R(q_2). \tag{1}$$

The align filter supplies a physical vehicle-to-body mount. In the Rust field layout this mount is stored in `q_bv0..q_bv3`, but its mathematical meaning is q_{bv}:

$$C_{bv} = R(q_{bv}), \qquad x_b = C_{bv} x_v, \qquad C_{vb} = C_{bv}^T, \qquad x_v = C_{vb} x_b. \tag{2}$$

The fusion facade exposes a single mount mode. In `MountMode::Auto`, Align seeds q_{bv} and the EKF continues estimating mount internally. In `MountMode::Manual`, the caller supplies a fixed q_{bv} and EKF mount states are frozen.

3 Nominal and Error States

The nominal state has 20 scalar components:

$$x = \begin{bmatrix} q_{nv} & v_n^T & p_n^T & b_g^T & b_a^T & q_{bv} \end{bmatrix}^T, \tag{3}$$

where q_{nv} rotates vehicle-frame vectors into NED, v_n is NED velocity, p_n is local NED position, b_g and b_a are additive gyro and accelerometer corrections in the raw IMU body frame, and q_{bv} is the physical vehicle-to-body mount.

The 18-dimensional error state is

$$\delta x = \begin{bmatrix} \delta\theta_v & \delta v_n & \delta p_n & \delta b_g & \delta b_a & \delta\psi_{bv} \end{bmatrix}^T. \tag{4}$$

The Rust index order is:

Indices	Error component
0..2	vehicle-attitude small angle $\delta\theta_v$
3..5	velocity error δv_n
6..8	position error δp_n
9..11	gyro-bias error δb_g
12..14	accelerometer-bias error δb_a
15..17	vehicle-to-body mount error $\delta\psi_{bv}$

4 Nominal Propagation

The EKF consumes one-sample raw body-frame IMU increments:

$$u_k = \{\Delta\theta_b, \Delta v_b, \Delta t\}. \tag{5}$$

Bias-corrected increments are

$$\Delta\theta_b^c = \Delta\theta_b - b_g\Delta t, \qquad \Delta v_b^c = \Delta v_b - b_a\Delta t. \tag{6}$$

The current mount rotates them into the vehicle frame:

$$\Delta\theta_v^c = C_{vb}\Delta\theta_b^c, \qquad \Delta v_v^c = C_{vb}\Delta v_b^c, \qquad C_{vb} = C_{bv}^T. \tag{7}$$

The generated nominal equations implement

$$q_{nv}^+ = \text{normalize}\left(q_{nv}^- \otimes \delta q(\Delta\theta_v^c)\right), \tag{8}$$

$$v_n^+ = v_n^- + C_{nv}\Delta v_v^c + g_n\Delta t, \qquad C_{nv} = R(q_{nv}^+), \tag{9}$$

$$p_n^+ = p_n^- + v_n^+\Delta t, \tag{10}$$

$$b_g^+ = b_g^-, \qquad b_a^+ = b_a^-, \qquad q_{bv}^+ = q_{bv}^-, \tag{11}$$

with $g_n = [0, 0, 9.80665]^T$ in NED.

5 Error-State Propagation

The covariance prediction uses generated discrete matrices:

$$\delta x_{k+1} = F_k\delta x_k + G_k w_k, \qquad P_{k+1}^- = F_k P_k^+ F_k^T + G_k Q_k G_k^T. \tag{12}$$

The 15-noise vector is

$$w = \begin{bmatrix} n_g & n_a & n_{bg} & n_{ba} & n_m \end{bmatrix}^T. \tag{13}$$

The diagonal covariance entries assembled at runtime are

$$Q_k = \text{diag}\left(\sigma_g^2\Delta t^2 I_3,\, \sigma_a^2\Delta t^2 I_3,\, \sigma_{bg}^2\Delta t I_3,\, \sigma_{ba}^2\Delta t I_3,\, \sigma_m^2\Delta t I_3\right). \tag{14}$$

If mount states are frozen, the mount random-walk block is set to zero and the mount covariance rows/columns are cleared.

6 Measurement Updates

Scalar observations use

$$r = z - h(\hat{x}), \tag{15}$$

$$S = HPH^T + R, \tag{16}$$

$$K = PH^T S^{-1}, \tag{17}$$

$$\delta x = Kr. \tag{18}$$

Covariance is updated with a scalar Joseph-form expression. The lateral and vertical NHC rows may be fused as one sequential two-row batch so the second row sees the covariance and accumulated correction from the first row.

6.1 GNSS Position and Velocity

GNSS position and velocity are local NED:

$$h_p(x) = p_n, \qquad H_p = \begin{bmatrix} 0 & 0 & I_3 & 0 & 0 & 0 \end{bmatrix}, \tag{19}$$

$$h_v(x) = v_n, \qquad H_v = \begin{bmatrix} 0 & I_3 & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{20}$$

The generated GNSS rows have no direct mount Jacobian. The runtime has optional mount update scales for GNSS rows; with the current generated H these only affect mount updates induced through covariance cross-correlation. The default public `fuse_gps` path passes zero scales.

6.2 Zero Velocity

The zero-velocity pseudo-measurement is a GNSS-velocity-shaped update:

$$z_v = 0, \qquad h_v(x) = v_n. \tag{21}$$

It does not directly observe mount because it is expressed in NED, not in vehicle coordinates.

6.3 Vehicle Speed and NHC

Predicted vehicle-frame velocity is

$$v_v = C_{nv}^T v_n. \tag{22}$$

The optional wheel/CAN speed measurement observes

$$h_x(x) = e_x^T v_v. \tag{23}$$

The non-holonomic constraints use zero lateral and vertical vehicle velocity:

$$h_y(x) = e_y^T v_v \approx 0, \qquad h_z(x) = e_z^T v_v \approx 0. \tag{24}$$

Because mount is already part of propagation, these instantaneous vehicle-speed rows have no direct mount Jacobian:

$$\frac{\partial(e_i^T v_v)}{\partial \delta\psi_{bv}} = 0, \qquad i \in \{x, y, z\}. \tag{25}$$

Mount is corrected through propagation-induced covariance with attitude and velocity.

6.4 Stationary Gravity

Stationary gravity updates use vehicle-frame accelerometer samples and update only x and y accelerometer channels:

$$f_v \approx -C_{nv}^T g_n + \eta. \tag{26}$$

The scalar rows are

$$h_x = e_x^T(-C_{nv}^T g_n), \qquad h_y = e_y^T(-C_{nv}^T g_n). \tag{27}$$

Before each stationary gravity scalar update, the runtime floors roll and pitch covariance to avoid over-confident attitude locking. The mount Jacobian block is zero for this pseudo-measurement.

7 Injection and Covariance Reset

After each update the error state is injected:

$$q_{nv}^{+} = q_{nv}^{-} \otimes \delta q(\delta\theta_v), \tag{28}$$
$$v_n^{+} = v_n^{-} + \delta v_n, \qquad p_n^{+} = p_n^{-} + \delta p_n, \tag{29}$$
$$b_g^{+} = b_g^{-} + \delta b_g, \qquad b_a^{+} = b_a^{-} + \delta b_a, \tag{30}$$
$$q_{bv}^{+} = \delta q(\delta\psi_{bv}) \otimes q_{bv}^{-}. \tag{31}$$

Both quaternions are normalized. The attitude and mount covariance blocks are reset with

$$G_r(\delta\theta) \simeq I - \frac{1}{2}\left[\delta\theta\right]_{\times}. \tag{32}$$

The complete reset matrix is block diagonal with G_r on the attitude and mount blocks and identity elsewhere:

$$P^{+} \leftarrow G_{\text{reset}} P^{+} G_{\text{reset}}^{T}. \tag{33}$$

8 Initialization and Mount Modes

At GNSS initialization the filter sets q_{nv} from GNSS heading or horizontal velocity course, copies GNSS NED position and velocity into p_n, v_n, sets biases to zero, and initializes q_{bv} from Align in auto mode or from the caller/reference source in manual mode. Covariance is diagonal with configuration-controlled bias and mount variances.

The fusion facade controls the mount mode:

- **Auto**: Align seeds q_{bv} when the EKF initializes; the EKF then estimates mount states internally.
- **Manual**: a supplied or reference q_{bv} is used as a fixed mount and EKF mount states are frozen.

9 Known Modeling Boundaries

- Mount affects inertial propagation directly by rotating raw IMU samples into the vehicle frame. This is the primary mount observability path.
- GNSS position, GNSS velocity, zero velocity, and stationary gravity do not directly observe q_{bv}.
- Lateral and vertical NHC observations can couple attitude, velocity, biases, and mount through covariance generated by propagation.
- The physical mount used for plotting is the stored q_{bv} itself.